# INTRODUCTION TO

a free, collaborative, multilingual knowledge base

WIKIDATA

# WHAT IS WIKIDATA?

A structured, open knowledge base supporting Wikipedia and other Wikimedia projects.

It stores linked data about people, places, events, concepts, and more.

Data is multilingual and editable by anyone, embracing a collaborative model

# WIKIDATA AND WIKIPEDIA

Wikidata acts as a centralized, structured knowledge base for Wikipedia and other Wikimedia projects.

Infobox templates in Wikipedia are auto-filled using live updates from Wikidata.

Interwiki links (interprojects or interlanguage) are managed centrally in Wikidata.

# HOW TO USE WIKIDATA?

Add or edit factual statements on items

Create new items for notable entities

**Use SPARQL to query the data via the Wikidata Query Service**

Embed data in Wikipedia and external websites

# DATA MODEL

Open, flexible graph structure for knowledge representation

A statement:

**Item** — has a **Property** — with a **Value**

Statements may have qualifiers (extra content or detail) and references (source).

# DATA MODEL

**Item** — has a **Property** — with a **Value**

- **Property** describes what kind of information is stated on the item, e.g., "date of birth," "occupation"
- **Value** is the actual data, e.g., "1867-11-07," "chemist".

| Item (Subject) | Property | Value (Object) | Explanation |
| --- | --- | --- | --- |
| Marie Curie (Q7186) | occupation (P106) | physicist (Q169470) | Marie Curie’s occupation includes physicist |
| Marie Curie (Q7186) | occupation (P106) | chemist (Q593644) | She also had the occupation of chemist |
| Marie Curie (Q7186) | award received (P166) | Nobel Prize in Physics (Q38104) | She received the Nobel Prize in Physics |
| Marie Curie (Q7186) | date of birth (P569) | 1867-11-07 | Her birthdate is November 7, 1867 |
| Marie Curie (Q7186) | place of birth (P19) | Warsaw (Q2692) | She was born in Warsaw |
| Marie Curie (Q7186) | educated at (P69) | University of Paris (Q35794) | She was educated at the University of Paris |
| Marie Curie (Q7186) | child (P40) | Irène Joliot-Curie (Q12345) | Her child Irène Joliot-Curie |

https://www.wikidata.org/wiki/Q7186

https://query.wikidata.org/

A public SPARQL endpoint for querying the entire Wikidata knowledge graph

# WHAT IS SPARQL?

- The standard query language for RDF data
- Allows querying by patterns of triples (subject-predicate-object)
- Supports filters, optional matches, aggregation, sorting, and more
- Used to query Wikidata's structured graph data

# BASIC STRUCTURE OF A WIKIDATA QUERY

- PREFIX declarations: shorthand for namespaces
- SELECT statement: specifies which variables (data) to return
- WHERE clause: graph pattern with triples and constraints

# EXAMPLE SPARQL QUERY

```
PREFIX wd: http://www.wikidata.org/entity/
PREFIX wdt: http://www.wikidata.org/prop/direct/

SELECT ?item ?itemLabel WHERE {
?item wdt:P31 wd:Q6256. # all countries
SERVICE wikibase:label { bd:serviceParam wikibase:language
"en". }
}
LIMIT 10
```

# USING THE WIKIDATA QUERY SERVICE INTERFACE

- Write or paste SPARQL queries into the editor
- Use “Add prefixes” button for common prefix declarations
- Click “Execute” to run the query and see results
- Save queries or generate short URLs for sharing
- Choose result views: table, graph, map, timeline, etc.

# EXAMPLE USE CASES

1. Retrieve all humans born in a specific country
2. Find all movies directed by a given person
3. Map geographic distribution of UNESCO World Heritage Sites
4. Analyze co-occurrence of concepts across items (items with multiple shared properties)

# EXAMPLE 1

```
SELECT ?person ?personLabel WHERE {
  ?person wdt:P31 wd:Q5;          # instance of human
          wdt:P19 ?place.         # place of birth
  ?place wdt:P17 wd:Q142.         # country is France (Q142)
  SERVICE wikibase:label { bd:serviceParam wikibase:language
"en". }
}
LIMIT 10
```

# EXAMPLE 2

```
SELECT ?movie ?movieLabel WHERE {
  ?movie wdt:P31 wd:Q11424;       # instance of film
         wdt:P57 wd:Q8877.        # director is Steven
Spielberg (Q8877)
  SERVICE wikibase:label { bd:serviceParam wikibase:language
"en". }
}
LIMIT 10
```

# EXAMPLE 3

```
SELECT ?landmark ?landmarkLabel ?coord WHERE {
  ?landmark wdt:P31 wd:Q9259;       # instance of World
Heritage Site
            wdt:P625 ?coord.         # coordinate location
  SERVICE wikibase:label { bd:serviceParam wikibase:language
"en". }
}
LIMIT 10
```

# EXAMPLE 4

```
SELECT ?person ?personLabel WHERE {
  ?person wdt:P31 wd:Q5;            # instance of human
          wdt:P106 wd:Q169470;      # occupation physicist
          wdt:P551 wd:Q64.          # residence Berlin (Q64)
  SERVICE wikibase:label { bd:serviceParam wikibase:language
"en". }
}
LIMIT 10
```